

Data Description (Grouped Data)

MEASURES OF CENTRAL TENDENCY

MEASURES OF VARIATION

MEASURES OF POSITION

Anthony S. Morfe_CAS_Math Cluster

Objective:

At the end of this presentation, students should be able to discuss data description for grouped data.



MEASURES OF CENTRAL TENDENCY-MEAN

$$\bar{x} = \frac{\sum fx}{n}$$

where: $\bar{x}$ = mean
f = frequency of each class
x = mid-interval value of each class *(midpoint or classmark)*
n = total frequency

Find the mean of the data.

Grade	Frequency	x
40 – 49	3	44.5
50 – 59	5	54.5
60 – 69	6	64.5
70 – 79	9	74.5
80 – 89	8	84.5
90 – 99	7	94.5

n = 38

Mean = $\frac{3(44.5) + 5(54.5) + 6(64.5) + 9(74.5) + 8(84.5) + 7(94.5)}{38}$

Mean = 73.71

MEASURES OF CENTRAL TENDENCY-MEDIAN

$$M_m = l + \left(\frac{\frac{n}{2} - cf}{f}\right)h$$

Where

l = lower class boundary of the median class

n = number of observations,

cf = cumulative frequency of class preceding the median class,

f = frequency of median class,

h = class size (assuming class size to be equal)

Find the location of the median:

n/2 = 38/2 = 19;

Between 19th and 20th data values

l = 69.5 n=38 cf=14 f=9 h=10

$$M_m = 69.5 + \left(\frac{\frac{38}{2} - 14}{9}\right)(10) = 75.06$$

Find the median of the data.

Grade	Frequency	<cf	
40 – 49	3	3	1st–3rd
50 – 59	5	8	4th–8th
60 – 69	6	14	9th–14th
70 – 79	9	23	15th–23rd
80 – 89	8	31	24th–31st
90 – 99	7	38	32nd–38th

Median Class

n = 38

MEASURES OF CENTRAL TENDENCY-MODE

$$M_o = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h$$

Where

l = lower class boundary of the modal class

h = size of the class interval (assuming all class sizes to be equal),

f_1 = frequency of the modal class,

f_0 = frequency of the class preceding the modal class,

f_2 = frequency of the class succeeding the modal class.

l = 69.5 h = 10 f_1 = 9 f_0 = 6 f_2 = 8

$$M_0 = 69.5 + \left(\frac{9 - 6}{2(9) - 6 - 8}\right)(10) = 77$$

Find the mode of the data.

Grade	Frequency	
40 – 49	3	
50 – 59	5	
60 – 69	6	
70 – 79	9	Modal Class
80 – 89	8	
90 – 99	7	

MEASURES OF VARIATION-
VARIANCE AND STANDARD DEVIATION

Sample Variance

$$s^2 = \frac{\sum f \cdot x^2 - \left(\sum f \cdot x\right)^2 / n}{n-1}$$

f = *frequency of each class*
x = *mid-interval value of each class (midpoint or classmark)*
n = *total frequency*

Find the sample SD of the data.

Grade	Frequency	x
40 – 49	3	44.5
50 – 59	5	54.5
60 – 69	6	64.5
70 – 79	9	74.5
80 – 89	8	84.5
90 – 99	7	94.5

n = 38

$\sum f \bullet x = 3(44.5) + 5(54.5) + 6(64.5) + 9(74.5) + 8(84.5) + 7(94.5) = 2801$

$\sum f \bullet x^2 = 3(44.5)^2 + 5(54.5)^2 + 6(64.5)^2 + 9(74.5)^2 + 8(84.5)^2 + 7(94.5)^2 = 215339.5$

$$Sample\ Variance = s^2 = \frac{215339.5 - \frac{(2801)^2}{38}}{38-1} = 239.9004267$$

$$Sample\ SD = s = \sqrt{s^2} = \sqrt{239.9004267} = 15.49$$

MEASURES OF POSITION
-PERCENTILE, DECILE, QUARTILE

$$P_k = l + \left(\frac{\frac{kn}{100} - cf}{f}\right)h$$

Where

l = lower class boundary of the percentile class

n = number of observations,

cf = cumulative frequency of class preceding the percentile class

f = frequency of percentile class

h = class size (assuming class size to be equal)

Find the location of P_{35}:

kn/100 = 35(38)/100 = 13.3;

13th data value

l = 59.5 n=38 cf=8 f=6 h=10

$$P_{35} = 59.5 + \left(\frac{\frac{35(38)}{100} - 8}{6}\right)(10) = 68.33$$

Find the P_{35} and D_7 of the data.

Grade	Frequency	<cf	
40 – 49	3	3	1st–3rd
50 – 59	5	8	4th–8th
60 – 69	6	14	9th–14th
70 – 79	9	23	15th–23rd
80 – 89	8	31	24th–31st
90 – 99	7	38	32nd–38th

Percentile Class

n = 38

MEASURES OF POSITION
-PERCENTILE, DECILE, QUARTILE

$$P_k = l + \left(\frac{\frac{kn}{100} - cf}{f}\right)h$$

Where

l = lower class boundary of the percentile class

n = number of observations,

cf = cumulative frequency of class preceding the percentile class

f = frequency of percentile class

h = class size (assuming class size to be equal)

Note: $D_7 = P_{70}$

Find the location of P_{70}:

kn/100 = 70(38)/100 = 26.6;

27th data value

l = 79.5 n=38 cf=23 f=8 h=10

$$D_7 = P_{70} = 79.5 + \left(\frac{\frac{70(38)}{100} - 23}{8}\right)(10) = 84$$

Find the P_{35} and D_7 of the data.

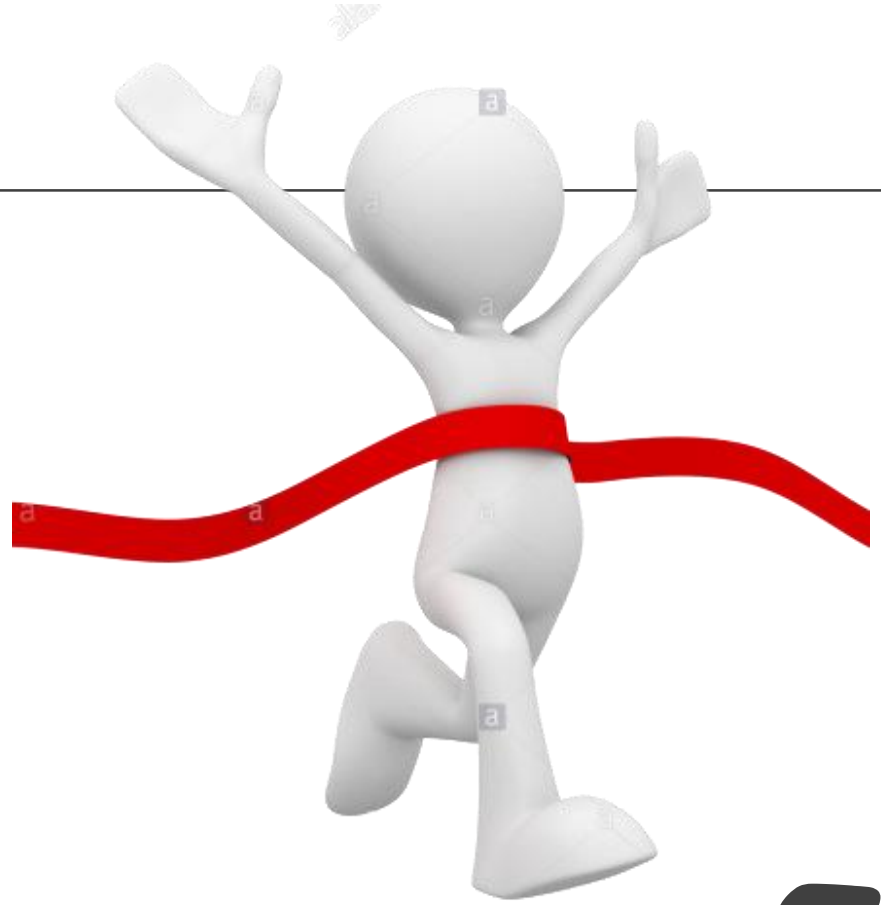
Grade	Frequency	<cf	
40 – 49	3	3	1st–3rd
50 – 59	5	8	4th–8th
60 – 69	6	14	9th–14th
70 – 79	9	23	15th–23rd
80 – 89	8	31	24th–31st
90 – 99	7	38	32nd–38th

Percentile Class

n = 38

Reference:

Bluman, A. (2012). *Elementary Statistics: A Step by Step Approach*,8e. McGraw-Hill Higher Ed.



Thank You!!!

CC BY NC